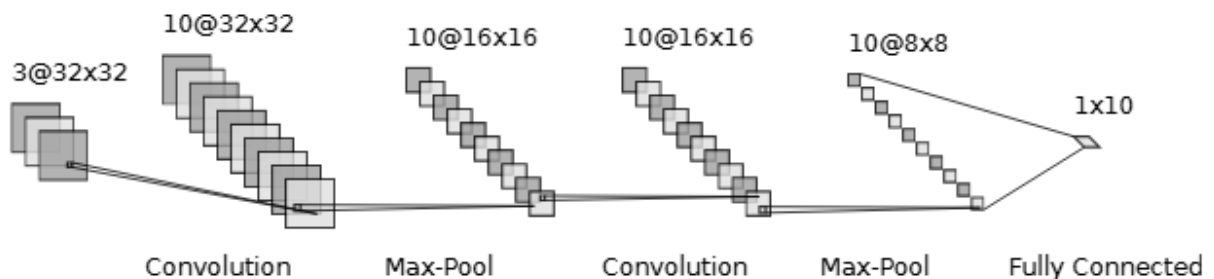


Q. Calculate the gradient vector at the pixel located at (1, 2). On the image, draw a vector starting from pixel (1, 2) that points roughly in the same direction as the gradient at that location.

10	10	10	10	10
10	20	10	10	10
10	30	40	10	10
10	10	10	10	10
10	10	10	10	10

Q. As we discussed in class, neural networks use non-linear activation functions to connect their hidden layers. Suppose a network uses a linear activation function in its hidden layers. What impact would this have on the expressive power of the network, regardless of the number of hidden layers? Explain briefly.

Q. Consider the convolutional neural network illustrated below, which performs image classification. The network accepts a 32×32 RGB image as input and produces a length-10 output vector. Its layers alternate convolution and max-pooling operations, followed by a single fully connected layer.



a. Assuming the convolution layers use stride = 1 and padding = 2, what is the convolutional kernel size shown in the Figure?

- b.** Assuming the max-pooling layers use stride = 2 and padding = 0, what is the pooling kernel size?
- c.** Calculate the total number of learnable parameters (weights and biases) in this network. Show your calculations and explain each step.